Recognition of Connected Components

Finding the connected components in the binary color image. A connected component is a set of black pixels in the image in which each black pixel in the component is a neighbor to at least one other pixel in the component. None of the pixels in one component are neighbors of pixels in any other component.

Assisting in the search for the components are two data structures: a one-dimensional array of linked lists called complists, and a two-dimensional array of integers, called compgrid. Each linked list in complists stores the coordinates of every pixel of a unique connected component. Initially, all the linked lists are empty. The size of the array complists is set to the maximum possible number of components that can exist in an image which is possible if every component is smallest in size, made up of a single pixel. Arraignment of these single pixel components in the image is as follows. Every odd numbered row contains components. Every even numbered row contains background (white) colored pixels in order to separate components in neighboring rows. The integer variable, rows, contains the number of rows in the image. Hence, the expression ((int) (rows / 2.0 + 0.5)) yields the number of rows in the image having components. Within an odd numbered row, every odd numbered pixel is a component and every even numbered pixel is white in order to separate neighboring components within the row. The integer variable, cols, contains the number of columns in the image. Hence, the expression ((int) (cols / 2.0 + 0.5)) yields the number of components within an odd numbered row. Therefore, the size of complists, called numlists, computes as [((int) (rows / 2.0 + 0.5)) * ((int) (cols / 2.0 + 0.5))].

The array compgrid keeps track of the component assigned to each pixel in the image. The size of compgrid is (rows x cols) having the same number of rows and columns as the image. Each index in the array complists denotes a particular component. The indices of complists range from 0 to (numlists-1). The value of -1, called NO_COMP, specifies that a pixel does not belong to any component. The initial value of all elements of array compgrid is NO_COMP.

The connected component recognition algorithm examines each pixel in the image one at a time starting with the pixel at coordinates (0, 0), row 0 and column 0 in the upper left hand corner of the image. Assume that the current pixel is the pixel at coordinates (i, j). If the color of the current pixel is white, the value of the element in compgrid corresponding to the current pixel remains as NO_COMP. If the color of the current pixel is black, the coordinates of the current pixel pigeonholes into the appropriate component's linked list in complists. For the current pixel, the only neighboring pixels processed already are the four pixels at coordinates (i-1, j-1), (i-1, j), (i-1, j+1) and (i, j-1), see figure 1. The algorithm retrieves from compgrid the component number for each of these four pixels discarding the NO_COMP values, white pixel neighbors. The current pixel belongs to one of the remaining components. The next actions taken depend on the number of remaining unique component numbers of these four neighbors.

(i-1, j-1)	(i, j-1)	(i+1, j-1)
(i-1, j)	(i, j)	

Figure 1 - The coordinates of the processed neighbors of the pixel (i, j).

If the number of remaining unique component numbers is zero, the current pixel (i, j) forms a new component. The algorithm inserts the coordinates of the current pixel (i, j) into the first empty linked list in complists thereby forming the new component. Assign the current pixel's new component number to the current pixel's corresponding element in compgrid.

If the number of remaining unique component numbers is one, the current pixel is a part of that one unique component. The algorithm inserts the coordinates of the current pixel (i, j) into the linked list of that component. Assign the current pixel's new component number to the current pixel's corresponding element in compgrid.

If the number of remaining unique component numbers is two or more, all of the unique components merge into the component with the smallest unique component number leaving the linked lists for the remaining unique components empty, for reuse, except the component with the smallest unique component number. The current pixel is a part of the merged component. The algorithm inserts the coordinates of the current pixel (i, j) into the linked list of the merged component. Assign the current pixel's new component number to the current pixel's corresponding element in compgrid. For each pixel in the merged component, ensure the pixel's corresponding element in compgrid is equal to the merged component's number.

Once finished with the current pixel (i, j), the algorithm moves on to the pixel (i+1, j) to the right of the current pixel. If the current pixel is the last pixel in the row, the algorithm moves to the first pixel in the next row. Once processing of all of the pixels in the image completes, each non-empty linked list of pixels in complists represents a unique connected component in the image.